

Lecture 6 Notes:

Formalism: Do Mathematical Statements Mean Anything?

Philosophy of Mathematics

Central Theme

Formalism is the family of views according to which mathematics is best understood through symbols, rules, axioms, and formal systems.

$$\boxed{\text{Mathematics} = \text{symbols} + \text{rules?}}$$

The guiding question:

$$\boxed{\text{Is mathematics about anything, or is it just rule-governed symbol manipulation?}}$$

Formalism tries to avoid metaphysical and epistemological problems by treating mathematics as formal activity rather than as knowledge of abstract objects.

1. The Formalist Impulse

Much mathematical practice looks like symbol manipulation according to rules.

For example, if we have:

$$a \times b = c,$$

then we may write:

$$b \times a = c.$$

And if we also have:

$$a \neq 0,$$

then we may write:

$$c/a = b.$$

The formalist asks whether this is not merely how mathematics is represented, but what mathematics essentially is.

Teaching Point

Formalism seizes on the rule-governed, symbolic character of mathematics and asks whether anything more is needed.

Whiteboard

Mathematics = symbols + rules for manipulating symbols

Do the symbols have to mean anything?

2. Why Formalism Is Tempting

Formalism becomes especially tempting when mathematics introduces entities that seem difficult to interpret.

Examples:

negative numbers

irrational numbers

transcendental numbers

imaginary numbers

points at infinity

Instead of asking what these entities really are, the formalist may say that mathematics only tells us how to use the corresponding symbols.

Teaching Point

Formalism offers a way to avoid metaphysical anxiety about mysterious mathematical objects.

Whiteboard

What is i ?

Formalist answer: a symbol governed by rules, e.g. $i^2 = -1$.

3. Term Formalism

The first version of formalism is term formalism.

Mathematical objects are symbols or terms.
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On this view:

2 = the numeral “2”

$8 + 2i$ = the symbol “ $8 + 2i$ ”

So mathematics has a subject matter, but its subject matter is linguistic.

What is mathematics about? : symbols
What are numbers? : numerals or symbol-types
How do we know mathematics? : by knowing rules for symbols

Teaching Point

Term formalism does not say that mathematics is about nothing. It says that mathematics is about its own signs.

Whiteboard

Term Formalism

numbers = numerals

mathematical objects = mathematical terms

4. Tokens and Types

Term formalism immediately faces a problem.

Consider:

$$0 = 0.$$

This cannot mean that one physical mark is identical to another physical mark. The two marks are distinct pieces of ink, chalk, or pixels.

So the term formalist may shift from tokens to types.

Token	Type
particular physical mark	repeatable abstract form
this ink mark	the symbol-form “0”
created and destroyed	shared by many tokens

But types are abstract objects. So term formalism may reintroduce the very kind of abstract ontology it hoped to avoid.

Teaching Point

If numbers are symbol-types, then term formalism still owes us an account of abstract objects.

Whiteboard

$$0 = 0$$

token identity? No.

type identity? Maybe.

But types are abstract.

5. Frege's Critique of Term Formalism

Frege presses the problem with ordinary equations.

Consider:

$$5 + 7 = 6 + 6.$$

If mathematical terms denote themselves, then this says:

$$“5 + 7” = “6 + 6”.$$

But that is false. The symbols are different.

So the term formalist may reinterpret equality as intersubstitutability:

$$“5 + 7” \text{ may be substituted for } “6 + 6”.$$

But then equality no longer means identity.

Teaching Point

Term formalism cannot preserve the ordinary meaning of mathematical equations.

Whiteboard

$$5 + 7 = 6 + 6$$

$$\text{Not: } “5 + 7” = “6 + 6”$$

$$\text{Maybe: } “5 + 7” \leftrightarrow “6 + 6” \text{ by rule}$$

But then $=$ has changed meaning.

6. Real Numbers and General Theorems

Term formalism has special trouble with real numbers.

Many real numbers have no finite name. One might try to identify π with its decimal expansion:

$$3.14159\dots$$

But the decimal expansion is infinite, not an ordinary concrete symbol.

If the term formalist appeals to limits of finite decimals, then the view begins to look like ordinary real analysis rather than formalism.

Term formalism also struggles with general mathematical theorems:

There are infinitely many primes.

Every continuous function on a closed interval is bounded.

Every nonconstant polynomial over $\mathbb{C}$ has a root.

These do not seem to be merely about symbols.

Teaching Point

Term formalism may handle simple calculations, but it struggles with higher mathematics and general mathematical propositions.

Whiteboard

$$\pi = 3.14159\dots$$

Is this a symbol?

What about theorems about all real numbers?

7. Game Formalism

The second version of formalism is game formalism.

Mathematics is like chess: a game played with symbols according to rules.

On this view, mathematical symbols are either meaningless or their meaning is irrelevant.

What is mathematics about? : nothing

What are numbers? : nothing, or no concern

What is mathematical knowledge? : knowledge of legal moves

A theorem is like a reachable position in a game.

axioms $\longrightarrow$ legal moves $\longrightarrow$ theorem

Teaching Point

Game formalism avoids ontology by denying that mathematics needs a subject matter.

Whiteboard

Game Formalism

axioms + rules = legal derivations

mathematical theorem = reachable position

8. Frege's Critique of Game Formalism

Frege's main objection is the problem of applicability.

Chess is rule-governed, but no one expects chess to explain planetary motion, engineering, electricity, or quantum mechanics.

Mathematics, by contrast, is deeply applicable to nature.

Chess	Mathematics
rule-governed	rule-governed
no scientific application expected	scientifically indispensable
no external truth needed	seems to express truth

If mathematics is only a meaningless game, why does it work so well?

Teaching Point

Frege argues that applicability requires meaning. Arithmetic can be applied because its equations express thoughts.

Whiteboard

If mathematics is like chess,

why does mathematics apply to science?

Meaning seems needed for application.

9. Instrumentalism Analogy

Game formalism resembles instrumentalism in philosophy of science.

Instrumentalism says that theories about unobservable entities are instruments for prediction, not literal descriptions of reality.

Likewise, game formalism says that mathematics is an instrument made of formal symbols, not a description of mathematical objects.

But both views face a similar question:

Why does the instrument work?

Teaching Point

Formalism may avoid ontology, but it inherits a serious explanatory burden: why is the formal instrument so effective?

Whiteboard

Instrumentalism: science as prediction-tool

Formalism: mathematics as symbol-tool

Question: why does it work?

10. Deductivism

Deductivism is a more sophisticated formalist view.

Mathematics studies what follows from axioms.

A theorem Φ is not simply asserted as true. Rather, the claim is:

If the axioms are true, then Φ is true.

This is why deductivism is sometimes called:

if-then-ism.

Deductivism preserves the importance of logical consequence. Rules of inference are not arbitrary moves; they must preserve truth under interpretation.

Teaching Point

Deductivism treats mathematics as the study of logical consequence from uninterpreted or variably interpreted axioms.

Whiteboard

Deductivism

$$A_1, \dots, A_n \vdash \Phi$$

means: if $A_1, \dots, A_n$ hold, then Φ holds.

11. Logical and Non-Logical Terms

Deductivism distinguishes logical terms from mathematical primitives.

Logical terms	Mathematical terms
and	point
or	line
not	number
if...then	set
for all	function
there exists	plane

The logical vocabulary keeps its meaning. The mathematical vocabulary may be left uninterpreted.

Teaching Point

For the deductivist, mathematics does not require fixed meanings for its primitive mathematical terms.

Whiteboard

Keep logic fixed.

Let mathematical primitives vary.

Study consequence.

12. Hilbert's Geometry

Hilbert's *Grundlagen der Geometrie* is the central example of deductivist mathematics.

Geometry can be axiomatized without relying on spatial intuition.

Once the axioms are stated, proofs should depend only on formal relations among the terms.

The famous idea:

“points,” “lines,” and “planes” could be “tables,” “chairs,” and “beer mugs.”

What matters is not the intuitive meaning of the words. What matters is whether the objects satisfy the axioms.

Teaching Point

Hilbert transforms geometry from the study of intuitive space into the study of abstract relational structure.

Whiteboard

point, line, plane

⇓

table, chair, beer mug

Same theory if the axioms are satisfied.

13. Implicit Definition

Hilbert's axioms do not define individual terms one at a time.

Instead, they define the primitive terms together by fixing their relations.

This is called implicit definition.

Axioms implicitly define a structure.

The primitive terms get their mathematical role from the axioms.

Teaching Point

Axioms can function as definitions, not by giving synonyms, but by fixing structural roles.

Whiteboard

Explicit definition: define one term by already understood terms.

Implicit definition: fix several terms by their relations.

Axioms define roles.

14. Frege versus Hilbert

Frege objects to Hilbert's use of axioms as definitions.

For Frege:

definitions give meanings

axioms state truths

So axioms cannot also be definitions.

Frege's dilemma:

Terms already have meaning	Terms lack meaning
axioms can be true or false but are not definitions	axioms cannot be true or false so they are not genuine axioms

Hilbert replies that this misunderstands modern axiomatics. A mathematical theory fixes primitive notions through their relations to each other.

Teaching Point

Frege wants meanings first and axioms second. Hilbert lets axioms fix the roles of the terms.

Whiteboard

Frege	Hilbert
meaning before axioms	axioms fix meaning/role
truth before consistency	consistency as legitimacy
concepts determine objects	structures determine positions

15. Consistency as Legitimacy

Hilbert famously claims:

If the axioms are consistent, then the things defined by them exist.

A more cautious formulation is:

Consistency is enough for mathematical legitimacy.

This is a crucial formalist idea. Mathematics need not first identify an independently given subject matter. It can set up a consistent axiomatic system and study its consequences.

Teaching Point

Hilbert replaces the question “What are these objects?” with the question “Are the axioms consistent?”

Whiteboard

Existence? $\longrightarrow$ Consistency

$\text{Con}(T) \Rightarrow$ legitimate theory T

16. Meta-Mathematics

Once formal systems are rigorously specified, they can be studied mathematically.

This is meta-mathematics.

Meta-mathematics = mathematics about formal mathematics

Meta-mathematical questions include:

Is this axiom system consistent?

Is this axiom independent of the others?

Is this formula derivable from these axioms?

Hilbert’s geometry already uses meta-mathematical methods by constructing models to show consistency and independence.

Teaching Point

Formalism helped create meta-mathematics: the mathematical study of formal systems themselves.

Whiteboard

Mathematics $\longrightarrow$ formal system

Formal system $\longrightarrow$ object of study

meta-mathematics

17. The Foundational Crisis

At the turn of the twentieth century, mathematics faced both enormous progress and serious foundational anxiety.

Progress:

real analysis

complex analysis

Cantor's set theory

the mathematics of the infinite

Crisis:

Russell's paradox

set-theoretic antinomies

doubts about infinite totalities

restrictions proposed by critics

Hilbert wanted to preserve classical mathematics without surrendering to skepticism.

Teaching Point

Hilbert's mature program is a defense of classical mathematics against paradox and restriction.

Whiteboard

Power of the infinite + paradoxes = foundational crisis

Hilbert: defend classical mathematics.

18. Hilbert's Program

Hilbert's goal:

Establish once and for all the certainty of mathematical methods.

His famous declaration:

No one shall drive us out of Cantor's paradise.

The strategy is to formalize ideal mathematics and then prove, using secure finitary methods, that these formal systems are consistent.

formalize $\longrightarrow$ prove consistency $\longrightarrow$ secure ideal mathematics

Teaching Point

Hilbert does not reject modern mathematics. He tries to justify it by proving that it is safe.

Whiteboard

Classical mathematics

$\Downarrow$

formal system T

$\Downarrow$

finitary proof of $\text{Con}(T)$

19. Finitary Mathematics

Hilbert distinguishes finitary mathematics from ideal mathematics.

Finitary mathematics is meaningful, secure, and directly surveyable.

It includes concrete calculations:

$$2 + 3 = 5$$

$$12553 + 2477 = 15030$$

It also includes bounded quantifiers:

$$\exists p < 101! + 2 \text{ such that } p \text{ is prime.}$$

This is finitary because only finitely many cases are involved.

Teaching Point

Finitary mathematics is Hilbert's secure base: meaningful mathematics concerning concrete symbolic constructions.

Whiteboard

Finitary

specific calculations

bounded search

directly checkable in principle

20. Bounded and Unbounded Quantifiers

Hilbert treats bounded and unbounded quantification differently.

Bounded quantifier	Unbounded quantifier
finite search	infinite range
finitary	ideal or non-finitary
effectively checkable in principle	not reducible to finite checking

Example of a bounded quantifier:

$$\exists p < 101! + 2 \text{ such that } p \text{ is prime.}$$

Example of an unbounded quantifier:

$$\exists p > 100 \text{ such that } p \text{ and } p + 2 \text{ are prime.}$$

The second statement ranges over infinitely many possible numbers.

Teaching Point

The finite/infinite distinction becomes epistemologically central for Hilbert.

Whiteboard

$$\exists p < N \quad \text{bounded}$$

$$\exists p \quad \text{unbounded}$$

finite check vs. infinite range

21. Stroke Symbols

Hilbert identifies the natural numbers with stroke-symbols.

$|, \quad ||, \quad |||, \quad ||||, \dots$

The symbol 2 abbreviates:

$||$

The symbol 3 abbreviates:

$|||$

The statement:

$3 > 2$

means that the stroke-symbol for 3 is longer than the stroke-symbol for 2.

These are not ordinary ink marks. They are symbol-types or directly surveyable symbolic forms.

Teaching Point

Hilbert's finitism has a Kantian element: it rests on immediately surveyable symbolic constructions.

Whiteboard

$1 = |$

$2 = ||$

$3 = |||$

$3 > 2$ because $|||$ extends $||$

22. Ideal Mathematics

Everything beyond finitary mathematics is ideal mathematics.

Examples:

unbounded number theory

real analysis

complex analysis

functional analysis

set theory

Ideal mathematics may not have direct meaning in the way finitary mathematics does. Rather, it is treated instrumentally, as a formal device that helps us reason efficiently.

Hilbert compares ideal mathematics to ideal points at infinity in geometry.

Teaching Point

Ideal mathematics is not rejected. It is justified as a useful formal extension of finitary mathematics.

Whiteboard

Finitary Mathematics	Ideal Mathematics
meaningful	formal/instrumental
secure	requires justification
finite symbols	infinity, analysis, set theory

23. Conservativeness and Consistency

Ideal mathematics is legitimate only if it does not lead to false finitary results.

This is the idea of conservativeness.

Ideal mathematics should be conservative over finitary mathematics.

In practice, this becomes the demand for consistency.

Prove $\text{Con}(T)$ by finitary means.

If an ideal theory T is consistent, then it cannot derive contradiction. If it is also conservative over finitary mathematics, then it cannot corrupt the meaningful base.

Teaching Point

Hilbert's formalism justifies ideal mathematics by proving that it is safe for finitary mathematics.

Whiteboard

Ideal theory T

safe if

$T \not\vdash 0 \neq 0$

$\text{Con}(T)$

24. Von Neumann's Summary

Von Neumann summarizes the Hilbert program in four stages:

1. Enumerate all symbols used in mathematics and logic.
2. Characterize the meaningful formulas.
3. Give a construction procedure for all provable formulas.
4. Prove, by finitary means, that the system proves the correct finitary statements.

The first three stages concern formalization.

The fourth is the crucial epistemic stage: proving that the formalized theory is safe.

Teaching Point

Hilbert's program is not merely formalization. It is formalization plus finitary justification.

Whiteboard

symbols $\rightarrow$ formulas $\rightarrow$ proofs $\rightarrow$ finitary consistency/safety proof

25. Gödel's First Incompleteness Theorem

Gödel's first incompleteness theorem strikes at the hope for complete formalization.

Let T be a sufficiently strong, consistent, effective formal system.

Then there is a sentence G such that:

$$T \not\vdash G$$

and, under suitable consistency assumptions,

$$T \not\vdash \neg G.$$

Roughly, G says:

“ G is not provable in T .”

If T is consistent, then G is true but not provable in T .

Teaching Point

No sufficiently strong consistent effective formal system captures all arithmetic truth.

Whiteboard

T consistent

$\Rightarrow$

$\exists G(T \not\vdash G \text{ and } T \not\vdash \neg G)$

Truth outruns formal proof.

26. Gödel's Second Incompleteness Theorem

Gödel's second incompleteness theorem is more directly damaging to the Hilbert program.

No sufficiently strong consistent system can prove its own consistency.

$T \not\vdash \text{Con}(T)$

assuming T is consistent and sufficiently strong.

Hilbert wanted finitary consistency proofs for ideal mathematical theories.

But if the ideal theory contains enough arithmetic, then its consistency cannot be proved inside the theory itself, much less inside a weaker finitary subsystem.

Teaching Point

The second incompleteness theorem blocks the original Hilbert program's hoped-for method of securing classical mathematics.

Whiteboard

Hilbert wants: finitary proof of $\text{Con}(T)$

Gödel shows: $T \not\vdash \text{Con}(T)$

if T is consistent.

27. What Gödel Does and Does Not Show

Gödel's theorems do not show that mathematics is inconsistent.

They do not show that formal systems are useless.

They do not show that formalization is unimportant.

But they do show that Hilbert's original epistemic ambition cannot be fulfilled in the simple way he hoped.

The Hilbert program cannot succeed in its original form.

Teaching Point

Gödel limits formalism; he does not abolish formal methods.

Whiteboard

Not: mathematics collapses.

Not: formal systems are useless.

But: formalism cannot secure everything from within.

28. Post-Gödel Options

The chapter notes two possible responses after Gödel.

First, one might challenge the formalization of consistency. There are different ways to express consistency, and some do not fall under the exact second incompleteness result.

Second, one might argue that finitary reasoning is inherently informal and cannot be fully captured by any one formal system.

Finitary reasoning $\neq$ one formal theory

These responses do not restore Hilbert's program in its original form, but they show that the philosophical situation is subtle.

Teaching Point

The failure of the original Hilbert program leaves room for weaker or revised formalist projects.

Whiteboard

Option 1: different consistency notions

Option 2: finitary reasoning is informal

Revised Hilbert programs?

29. Curry's Formalism

The chapter ends with Haskell Curry.

Curry's central claim:

Mathematics is the science of formal systems.

This is different from crude game formalism. Mathematical assertions are not merely moves within a system. They are often statements about formal systems.

A typical mathematical assertion becomes:

Φ is a theorem of T .

Curry therefore treats mathematics as objective, but its subject matter is formal systems themselves.

Teaching Point

Curry turns formalism into meta-mathematics: mathematics studies formal systems.

Whiteboard

Mathematics = science of formal systems

$$\Phi \in \text{Th}(T)$$

Φ is a theorem of T

30. Curry on Consistency and Usefulness

Curry does not follow Hilbert in requiring consistency proofs before accepting a formal system.

For Curry, acceptability is pragmatic.

Criteria include:

intuitive evidence of premises

consistency

usefulness

A consistency proof is useful, but it is not always necessary for accepting a system. If an inconsistency appears, the system may be modified rather than abandoned completely.

Teaching Point

Curry's formalism is less vulnerable to Gödel because it does not require Hilbert-style finitary consistency proofs.

Whiteboard

Acceptability = purpose-relative

consistency + usefulness + evidence

31. The Problem of Informal Mathematics

Curry’s view faces a problem.
If mathematics is the science of formal systems, what about mathematics before formalization?
Were the following people not doing mathematics?

Archimedes

Fermat

Euler

Cauchy

Even today, much mathematical practice is informal. Mathematicians do not usually write their work in fully formal deductive systems.

Teaching Point

Formalism risks mistaking the formal reconstruction of mathematics for mathematics itself.

Whiteboard

Formalized mathematics $\neq$ all mathematical practice

Does formalism lose the living practice of mathematics?

Conclusion: The Formalist Family

Formalism is not one view. It is a family of views united by emphasis on symbols, rules, axioms, formal systems, and derivations.

Term	Formalism	Game Formalism	Deductivism	Hilbert Program
	math about symbols	math as symbol game	math as consequence	ideal math justified by consistency
	objects = terms	no subject matter	if axioms, then theorem	finitary proof of safety
	Frege: equations fail	Frege: application fails	meaning thinned out	Gödel problem

Curry adds a later version:

Mathematics is the science of formal systems.

Final Framing

To teach this chapter well, present formalism as both powerful and limited.

It is powerful because it clarifies proof, rigor, axioms, formal systems, consistency, and meta-mathematics.

It is limited because mathematics appears to involve meaning, application, informal understanding, and truth that cannot be reduced to formal derivability.

Formalism reveals the power of symbols, but struggles with meaning and application.

Gödel shows that formal systems cannot fully contain arithmetic truth.

Discussion Questions

1. What motivates formalism as a philosophy of mathematics?
2. Is mathematics really just symbol manipulation?
3. What is the difference between term formalism and game formalism?
4. Can numbers be identified with numerals?
5. Why does Frege think formalism cannot explain applicability?
6. Is mathematics more like chess or more like science?
7. What does deductivism add to simple game formalism?
8. Can axioms function as definitions?
9. Was Frege or Hilbert right about the relation between definitions and axioms?
10. Is consistency enough for mathematical existence?
11. What is the difference between finitary and ideal mathematics?
12. Why did Hilbert want finitary consistency proofs?
13. What does Gödel's first incompleteness theorem show?
14. Why is the second incompleteness theorem a problem for Hilbert's program?
15. Is Curry right that mathematics is the science of formal systems?

Key Terms

Formalism A family of views emphasizing mathematics as symbol manipulation, formal rules, axioms, and formal systems.

Term formalism The view that mathematical objects are symbols or terms.

Game formalism The view that mathematics is like a game played with symbols according to rules.

Token A particular physical inscription or mark.

Type A repeatable abstract symbol-form shared by many tokens.

Deductivism The view that mathematics studies what follows logically from axioms.

If-then-ism Another name for deductivism; the view that mathematical theorems are conditional on axioms.

Implicit definition A definition in which axioms fix the roles of primitive terms by specifying their relations.

Consistency The property of a formal system that it does not derive a contradiction.

Meta-mathematics The mathematical study of formal mathematical systems.

Finitary mathematics For Hilbert, secure mathematics concerning directly surveyable finite symbolic constructions.

Ideal mathematics Mathematics involving ideal elements such as infinity, set theory, and analysis, justified instrumentally by consistency or conservativeness.

Conservativeness The property that an extension of a theory does not prove any new claims of a specified base kind, especially no false finitary claims.

Hilbert program Hilbert's project of formalizing ideal mathematics and proving its consistency by finitary means.

Gödel's first incompleteness theorem The result that any sufficiently strong, consistent, effective formal system is incomplete.

Gödel's second incompleteness theorem The result that no sufficiently strong consistent formal system can prove its own consistency.

Curry's formalism The view that mathematics is the science of formal systems.